2025

AUSTRALIAN MATHEMATICS COMPETITION

AMC.

AUSTRALIAN MATHS TRUST

Level C
Junior
Grades 8 - 9

DATE
27 Sept.

TIME ALLOWED
75 minutes

Information 考试信息

- There are 30 questions: 25 multiple-choice and 5 integer.
  本次考试共 30 题，包含 25 道选择题和 5 道答案为整数的填空题。
- You are allowed 75 minutes to do the questions. There is no extra reading time.
  考试时长为 75 分钟，无额外读题时间。
- There are no penalties for incorrect answers. You should attempt all questions.
  答错不扣分，鼓励学生尝试解答所有问题。
- The questions have been checked thoroughly. Each question is clearly written and no further explanation will be given.
  本场考试的所有问题已仔细核查。题目陈述清晰，不提供更多解释。
- Diagrams are not drawn to scale. They are intended only as aids.
  图形未按比例绘制，仅供参考。

Competition rules 考试规则

- You must maintain silence at all times.
  考试过程中请保持安静。
- Do not open the paper until told to do so.
  收到监考老师指示前，请勿打开试卷。
- Mobile phones and smart watches are not allowed. If you have one you must give it to a supervisor, who will store it in a safe place.
  禁止使用手机和智能手表。如有携带这些设备，请立刻交与监考老师保管。
- You may use printed language dictionaries.
  允许使用纸质语言词典。
- You may NOT use calculators.
  不允许使用计算器。
- You may not borrow equipment without a supervisor's permission.
  未经监考老师许可，禁止向他人借用工具。
- You must not leave your seat during the competition. You must remain seated until the competition finishes.
  考试期间禁止离开座位，直到考试结束。
- If you wish to leave the room, a supervisor must accompany you. Raise your hand and wait for a supervisor to come to you.
  如需离开考场，必须举手示意并等待监考老师陪同。
- If you have any other questions or problems, raise your hand and wait for a supervisor.
  如果有其他问题，请举手示意监考老师。

No student may sit the competition more than once or sit more than one division of the competition. AMT conducts integrity checks on competition results and reserves the right to withhold results or disqualify students if plagiarism or duplicate sittings are suspected.

IMPORTANT

All papers must be immediately returned to a supervisor at the end of the competition.
考试结束后，由监考老师回收所有试卷。

This paper is not for circulation or distribution until Wednesday 1 October 2025.
在 2025 年 10 月 1 日星期三之前禁止讨论或传播试卷内容。

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Junior Division

Questions 1 to 10, 3 marks each
1-10 题，每题 3 分

1. A speaker's volume dial is shown.

Which of the following is the best estimate of the current volume setting?

扬声器的音量调节旋钮如图所示，请问以下哪个选项最接近当前的音量设置？

(A) 5.6 (B) 6.7 (C) 7.1 (D) 7.9 (E) 8.3

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2. What is $20 \times 2 + 5$?

请问 $20 \times 2 + 5$ 的值是多少?

(A) 27 (B) 45 (C) 50 (D) 140 (E) 200

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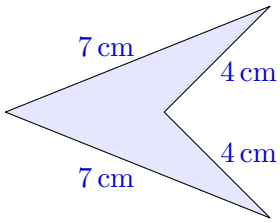
3. What is the perimeter of this dart?

请问这个飞镖的周长是多少？



(A) 18 cm (B) 19 cm (C) 20 cm (D) 21 cm (E) 22 cm

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4. Apples cost \$3 each. One apple and two bananas cost \$7. What is the cost of one banana?

苹果每个售价 3 美元。一个苹果和两根香蕉的售价为 7 美元。请问一根香蕉的售价是多少？

(A) 50c 50 美分 (B) \$1 1 美元 (C) \$2 2 美元

(D) \$3 3 美元 (E) \$4 4 美元

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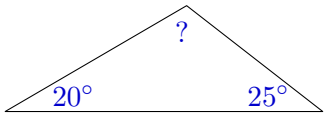
5. What is the value of the missing angle in the diagram?

请问图中缺失的角度值是多少？



(A) 45° (B) 75° (C) 105° (D) 135° (E) 165°

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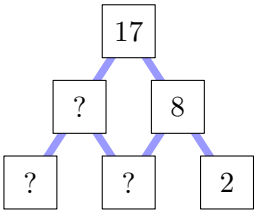
6. In this number puzzle, two adjacent numbers in the same row should add to the number written above them.

What is the sum of the three unknown numbers marked with question marks?

该数字谜题中，同一行相邻两数之和等于其上方的数字。请问标记问号处的三个未知数之和是多少？



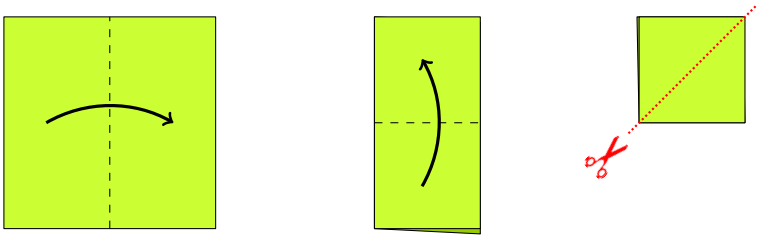
(A) 16 (B) 17 (C) 18 (D) 19 (E) 20

---

7. A paper square is to be folded twice and then cut along the dotted line as shown.

After this, how many pieces of paper are there?

如图所示，将一张正方形纸片对折两次后沿点线剪开。完成上述步骤后，请问这张纸片会被分成多少个部分？



(A) 2 (B) 4 (C) 8 (D) 12 (E) 16

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8. The number 20.25 expressed as a fraction is

请将数字 20.25 表示为分数形式。

(A) $\frac{41}{2}$ (B) $\frac{61}{3}$ (C) $\frac{81}{4}$ (D) $\frac{101}{5}$ (E) $\frac{121}{6}$

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9. What is the obtuse angle between the hands of a clock at 5 pm?

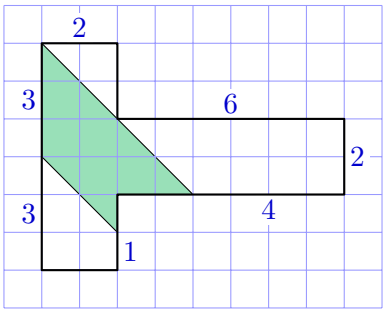
请问下午 5 点时，时钟的时针与分针之间的钝角是多少度？

(A) 30° (B) 90° (C) 120° (D) 135° (E) 150°

---

10. A T puzzle is made up of four pieces, shown here on a grid of 1 cm squares. What is the area of the shaded piece, in square centimetres?

如图所示，在一个由边长为 1 cm 的方格组成的网格上有一个由四块拼图碎片组成的 T 型拼图。请问阴影部分的面积是多少平方厘米？



(A) 4 (B) 6 (C) 8 (D) 10 (E) 12

Questions 11 to 20, 4 marks each
11-20 题，每题 4 分

11. Timmy has a bag with 9 balls in it, numbered 1 to 9. He pulls one ball out at random and looks at the number. Which of the following is the most likely to occur?

Timmy 的袋中有 9 颗球，编号从 1 至 9。他从中随机取出一颗球并查看上面的数字。请问以下哪种情况最有可能发生？

(A) The number is odd. 该数字为奇数

(B) The number is even. 该数字为偶数

(C) The number is less than 4. 该数字小于 4

(D) The number is more than 5. 该数字大于 5

(E) The number is a multiple of 3. 该数字是 3 的倍数

---

12. In the fraction $\frac{11}{\triangle}$, which positive integer should replace the triangle to make the fraction as close to 3 as possible?

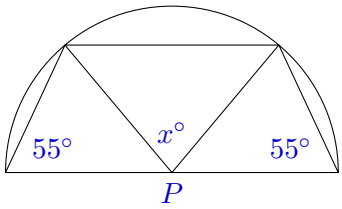
请问在分数 $\frac{11}{\triangle}$ 中，用哪个正整数替换其中的三角形，能使该分数尽可能接近 3 ？

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

---

13. In this diagram, P is the centre of the diameter of the semicircle. What is the value of x?

如图所示， P 是该半圆直径的中点。请问 x 的值是多少？



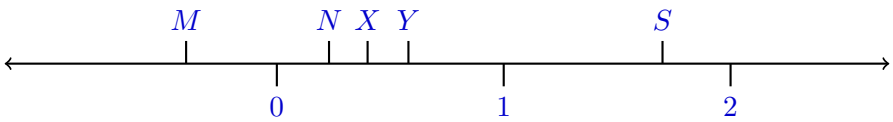
(A) 40 (B) 45 (C) 50 (D) 55 (E) 60

---

14. The fractions represented by the points X and Y are multiplied.

What point on the number line best represents their product?

将点 X 和 Y 所表示的分数相乘。请问数轴上的哪个点最能表示其乘积？



(A) M (B) N (C) X (D) Y (E) S

---

15. Nathan wrote down the list formed by repeating the digits of 2025:
Nathan 写下了一个由数字 2025 循环组成的数列：

$$2, 0, 2, 5, 2, 0, 2, 5, 2, 0, 2, 5, \ldots$$

He stopped when the total of all the numbers in his list reached 2025. How many numbers are in his list?
当数列中所有数字相加的总和达到 2025 时，他停止书写。请问这个数列中有多少个数字？

(A) 225 (B) 500 (C) 825 (D) 900 (E) 1000

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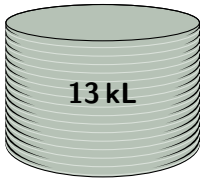
16. In early August, I have 13 000 litres of water stored in this tank.
I use 80 litres per day and don't refill it. When will it run out?
八月初，该水箱储水量为 13 000 升。若每日消耗 80 升且不进行补充，请问水箱中的水何时会用尽？



(A) early September 九月初
(B) late October 十月末
(C) early December 十二月初
(D) mid January 一月中旬
(E) late February 二月末

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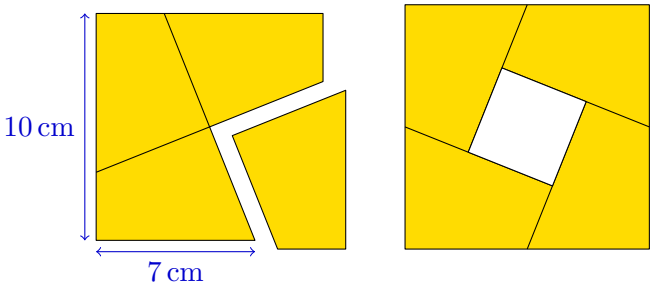
17. A 10 cm × 10 cm square is cut into four identical pieces, as shown on the left. The pieces are rearranged to form a larger square with a square hole in the middle.



一个 10 cm × 10 cm 的正方形被分割成四个相同的部分（如左图所示），重新排列这些部分后，形成一个中间有正方形孔洞的大正方形。
What, in square centimetres, is the area of the hole?
请问，这个正方形孔洞的面积是多少平方厘米？

(A) 4 (B) 8 (C) 10 (D) 12 (E) 16

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18. Alice and Bob are standing in a queue, with Alice somewhere in front of Bob. In the queue, there are 17 people behind Alice, 23 people in front of Bob, and 8 people between them. How many people are in the queue?
Alice 和 Bob 在排队，Alice 排在 Bob 前面。队伍中 Alice 后面有 17 人，Bob 前面有 23 人，他们之间有 8 人。请问该队伍中共有多少人？

(A) 30 (B) 31 (C) 32 (D) 33 (E) 34

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19. Kerri can walk the 9 km around the block at a steady rate in 1 hour and 45 minutes. She can run around the same block in 36 minutes. If Kerri walks for the first 3 km and runs for the rest, how long will it take for her to get around the block?
Kerri 能以稳定的速度走完一圈 9 km 的街区，用时 1 小时 45 分钟。她跑步绕行同一个街区，用时 36 分钟。如果 Kerri 先步行 3 km，剩余的路程跑步完成，请问她绕行整个街区需要多长时间？

(A) 58 min 58 分钟 (B) 59 min 59 分钟 (C) 60 min 60 分钟
(D) 61 min 61 分钟 (E) 62 min 62 分钟

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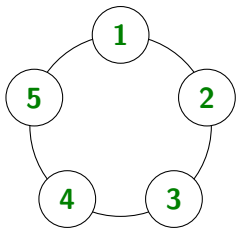
20. Starting at one of these digits, then moving all the way around the circle in either direction, a 5-digit number can be read off.
For instance 34512 and 43215 are 2 of the 10 possibilities.
When all 10 of these numbers are added, the result is a six-digit number. What is the largest digit of this number?



从图中任意一个数字出发，按顺时针或逆时针方向绕一圈，可以读出一个五位数。这样的五位数一共有 10 种可能，例如，34512 和 43215 就是其中的两种。将这 10 个所有可能的五位数相加，可以得到一个六位数。请问这个六位数中最大的数字是多少？

(A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Questions 21 to 25, 5 marks each
21-25 题，每题 5 分

21. A mini cross number puzzle is shown. No answer starts with 0.
一个迷你纵横数字谜题如图所示。所有答案均不以 0 开头。

Across 横向提示

1. A perfect cube
一个完全立方数

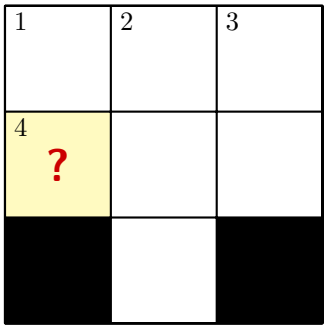
4. An odd square number
一个数既是奇数，也是平方数

Down 纵向提示

1. A number whose prime factors each have a single digit
一个所有质因数均为一位数的数字

2. A power of 2
一个 2 的幂

3. A multiple of 5
一个 5 的倍数

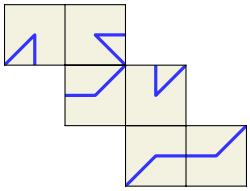


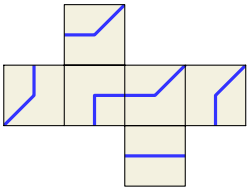
Which digit goes in the highlighted cell?
请问，高亮单元格中应该填入哪个数字？

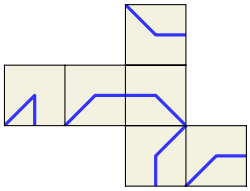
(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

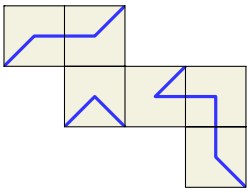
22. Max has drawn a *simple closed curve* on the surface of a cube: his pen traces a continuous path that ends up back where it started, without retracing or lifting off the cube. Which diagram could be a net for the cube?

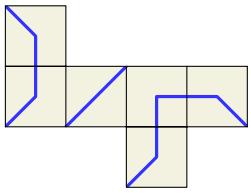
Max 在一个立方体表面画了一条简单闭合曲线：他的笔沿着一条连续路径移动，最终回到起点，且在移动过程中没有折回或离开立方体表面。请问以下哪个选项可能是该立方体的展开图？

(A) 

(B) 

(C) 

(D) 

(E) 

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23. My local hardware shop has screws in packs of 500, 200 and 50 screws. I manage to buy the number of screws I need for no more than \$100.

What is the most screws I could have bought?

本地五金店出售螺丝，包装规格分别为每包 500 个、每包 200 个以及每包 50 个。我计划以不超过 100 美元的价格买到所需数量的螺丝。

请问我最多能购买到多少个螺丝？

Screws in pack 螺丝包装规格	Price 价格
500	\$36
200	\$16
50	\$10

(A) 1000 (B) 1100 (C) 1200 (D) 1300 (E) 1350

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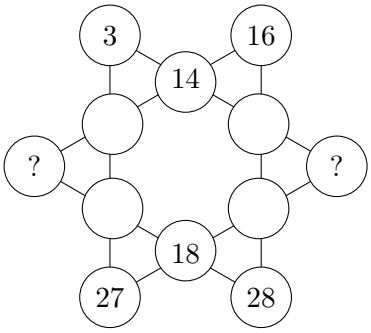
24. Arwen writes a number in each empty circle so that every straight line of four numbers adds to 60.

When the puzzle is complete, what is the sum of the two numbers in the leftmost and rightmost circles?

Arwen 在每个空白圆圈中填入一个数字，使得每条直线上的四个数之和均为 60。完成谜题后，请问最左端和最右端圆圈中的两数之和是多少？



(A) 28 (B) 29 (C) 30 (D) 31 (E) 32

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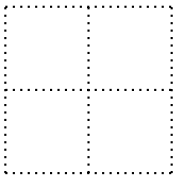
25. Tilly has a number of identically sized red, green and blue square tiles. She wishes to make a pattern on her wall using four of these tiles in a 2-by-2 grid. However, she doesn't want any two tiles that share a common edge to be the same colour. In how many ways can she make this pattern?



Tilly 有若干大小相同的红色、绿色和蓝色方形瓷砖。她想用四块这样的瓷砖在墙上拼出一个 2×2 的图案，但要求任意两块共边的相邻瓷砖颜色都不相同。请问她可以拼出多少种不同的图案？

(A) 6 (B) 9 (C) 12 (D) 18 (E) 24

For questions 26 to 30, shade the answer as an integer from 0 to 999 in the space provided on the answer sheet.

问题 26-30 的答案为 0-999 之间的整数，请将答案填涂在答题卡对应区域。

Questions 26 to 30 are worth 6, 7, 8, 9 and 10 marks, respectively.

第 26-30 题分别为 6, 7, 8, 9, 10 分。

26. Shaun constructed a large cube using small cubes of size $1 \times 1 \times 1$. Two small cubes are called *neighbouring* if they share a common face. So each small cube has at most six neighbouring cubes. It is known that 132 small cubes have exactly four neighbouring cubes each. How many small cubes have exactly five neighbouring cubes?

Shaun 用若干个 $1 \times 1 \times 1$ 的小立方体搭建一个大立方体。如果两个小立方体共用一个公共面，则称它们为*相邻立方体*。因此，每个小立方体最多有六个相邻立方体。已知有 132 个小立方体恰好每个都有四个相邻立方体。请问有多少个小立方体恰好有五个相邻立方体？

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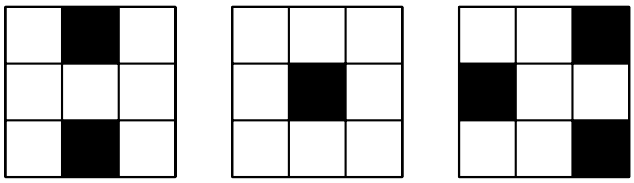
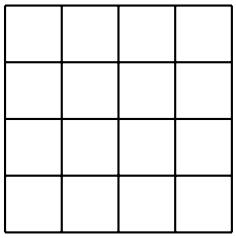
27. A grid is called *sparse* if in each 2-by-2 block, exactly 1 square is black and the rest are white. Three examples of sparse 3-by-3 grids are shown on the left:
若一个网格中的每个 2×2 子格中，恰好有 1 个黑色方格，其余均为白色方格，则称此网格为 稀疏网格。下图左侧展示了三个 3×3 稀疏网格示例：





Counting rotations and reflections as different, how many ways are there of filling the **4-by-4** grid to make a sparse grid?
在将旋转和镜像视为不同方案的情况下，请问有多少种方式可以填充一个 **4×4** 的网格以形成一个稀疏网格?

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28. Madeleine thinks of four whole numbers. She writes down all six possible sums of two of the numbers on a piece of paper. Sebastien can read the numbers 57, 72, 73, 79 and 88 on this piece of paper, but cannot read the sixth sum. What is the sixth sum that Madeleine wrote?
Madeleine 选了四个整数，并将所有可能的两数之和（六个）写在纸上。Sebastien 可以看到纸上的数字：57、72、73、79 和 88，但第六个和数看不清。请问 Madeleine 写的第六个和数是多少？

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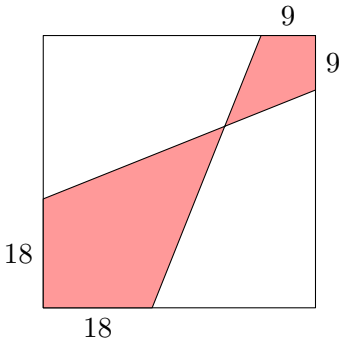
29. This square has sides of length 45 centimetres and an area of 2025 square centimetres.
What is the shaded area, in square centimetres?
已知该正方形的边长为 45 厘米，面积为 2025 平方厘米。请问图中阴影部分的面积为多少平方厘米？



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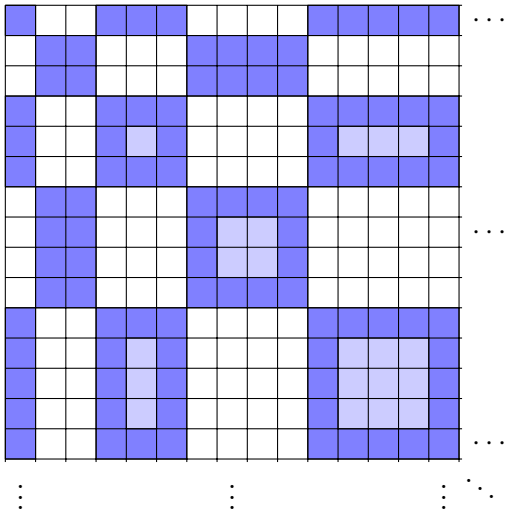
30. Martin made a mosaic with 2025 square tiles arranged in a 45-by-45 grid. The mosaic is a blue-and-white chequerboard of differently shaped rectangles. The top-left corner is shown here, including 25 rectangles.



The first row of rectangles is one tile high, the second two tiles high, the third three tiles high, and so on. The first column of rectangles is one tile wide, the second two tiles wide, the third three tiles wide, and so on. The bottom-right rectangle is a 9-by-9 square.

Martin 用 2025 块正方形瓷砖拼成一个 45×45 的网格状马赛克图案。这幅马赛克是一个蓝白相间的棋盘格图案，由不同形状的矩形组成。左上角的部分如图所示，包含 25 个矩形。第一行的矩形高 1 块瓷砖，第二行高 2 块瓷砖，第三行高 3 块瓷砖.....以此类推。第一列的矩形宽 1 块瓷砖，第二列宽 2 块瓷砖，第三列宽 3 块瓷砖.....以此类推。右下角的矩形是一个 9×9 的正方形。

Every blue rectangle has a 1-wide border of dark blue tiles, and the interior is composed of light blue tiles.
每个蓝色矩形都有一个宽 1 块瓷砖的深蓝色边框，且内部由浅蓝色瓷砖组成。

How many of the 2025 tiles in the mosaic are dark blue?
2025 块瓷砖拼成的马赛克图案中，请问有多少块瓷砖是深蓝色的？